








Advanced Time Series Analysis and Forecasting

Seminar 5: Bayesian VAR, factor models and nowcasting



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BBE**: Bernanke–Boivin–Eliasz (FAVAR, 2005)
- **BGR**: Bańbura–Giannone–Reichlin (large Bayesian VARs, 2010)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVAR**: Bayesian Vector AutoRegression
- **CPI**: Consumer Price Index
- **DFM**: Dynamic Factor Model
- **EM**: Expectation–Maximisation (algorithm)
- **ESI**: Economic Sentiment Indicator (European Commission)
- **ESS**: Effective Sample Size
- **FAVAR**: Factor-Augmented Vector AutoRegression
- **FFR**: Federal Funds Rate
- **FRED**: Federal Reserve Economic Data
- **FRED-MD**: FRED Monthly Database for macroeconomic research (McCracken–Ng)

Acronyms used in this material (2/2)

- **GDP**: Gross Domestic Product
- **GLP**: Giannone–Lenza–Primiceri (hierarchical prior selection, 2015)
- **HICP**: Harmonised Index of Consumer Prices
- **IC**: Interval de încredere (confidence interval)
- **IP**: Industrial Production
- **MCMC**: Markov Chain Monte Carlo
- **MEDIUM**: the 20-variable system of BGR
- **MIDAS**: MIxed DAta Sampling (regression)
- **NIW**: Normal–inverse-Wishart (prior)
- **OLS**: Ordinary Least Squares
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **RW**: Random Walk
- **SMALL**: the 3-variable system of BGR (employment, CPI, federal funds rate)
- **US**: United States
- **VAR**: Vector AutoRegression

Today's question and seminar structure

- The question of the seminar
  - **how do we estimate, forecast and nowcast with many series and few observations?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	conjugate updating and empirical Bayes; Minnesota prior variances and a ridge posterior	Solved, Proposed	A1
A3, A4	Gibbs conditionals and MCMC diagnostics; dummy observations	Solved, Proposed	A1, A3
A5, A6	principal components of an equicorrelated panel; Bai–Ng criteria	Solved, Proposed	A5
A7, A8	Mariano–Murasawa weights; news and MIDAS weights	Solved, Proposed	A7
B1, B2	hyperparameters by marginal likelihood (US); a Romanian BVAR	Solved, Proposed	B1
B3, B4	factors on three samples; FAVAR and the number of factors	Solved, Proposed	B3
B5, B6	nowcasting Romanian GDP; the news of September 2026	Solved, Proposed	B5
C1, C2	surveys after 2020; what is wrong in an AI answer?	Proposed	B5, A1–A8

- **[Solved]**: the full solution is in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it following the model, and the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions, including the critique of an AI answer

Data used

Series	Source	Frequency	Period
FRED-MD-format US panel (117 series)	FRED	monthly	1960–2026
RO IP, HICP, ROBOR 3M, unemployment, ESI	Eurostat	monthly	2005–2026
EUR/RON	BNR reference rate	monthly average	2005–2026
RO nowcasting panel (11 series), real GDP	Eurostat	monthly, quarterly	2003–2026

- In the notebook: `bvar_fit`, `glp_optimize`, `niw_logml`, `em_pca`, `bai_ng`, `favar_run`, `ro_vintage`, `bridge_nowcast`, `midas_nowcast`, `dfm_nowcast`, `news_decomposition`; no account or key is needed

Prerequisites for Today (1/22): Bayes' rule and the marginal likelihood

- Bayes' rule: the posterior is proportional to the likelihood times the prior

$$p(\theta \mid y) = \frac{p(y \mid \theta)\, p(\theta)}{p(y)} \propto p(y \mid \theta)\, p(\theta)$$

  - θ: the parameters; y: the data; $p(\theta)$: the prior; $p(y \mid \theta)$: the likelihood
- Marginal likelihood: the density of the data averaged over the prior

$$p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta$$

  - it does not depend on θ; it depends on the prior and its hyperparameters
  - a high $p(y)$: the prior predicted the data well; it is used to choose hyperparameters and to compare models
- Hyperparameters: the parameters of the prior itself (for example its variance); **empirical Bayes** sets them by maximising $p(y)$

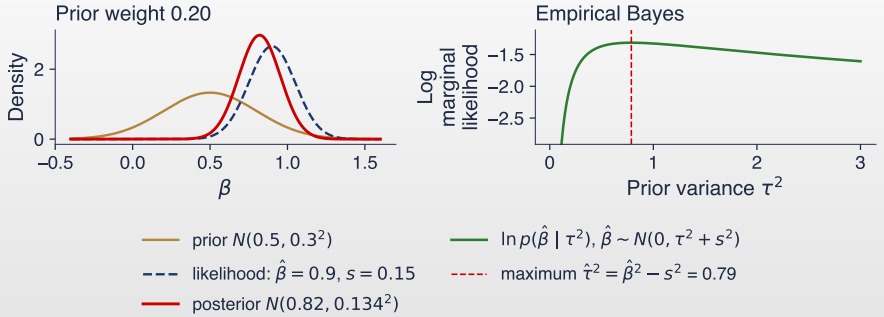
Prerequisites for Today (2/22): conjugate Normal updating

- Normal prior $\beta \sim N(m_0, v_0)$; data summarised by $\hat{\beta} \mid \beta \sim N(\beta, s^2)$
  - $\hat{\beta}$: the OLS estimate, with standard error s; m_0, v_0: prior mean and variance
- Posterior $\beta \mid y \sim N(m_1, v_1)$: precisions (inverse variances) add, means are precision-weighted

$$\frac{1}{v_1} = \frac{1}{v_0} + \frac{1}{s^2}, \qquad m_1 = v_1\left(\frac{m_0}{v_0} + \frac{\hat{\beta}}{s^2}\right)$$

  - prior weight v_1/v_0: between 0 (vague prior) and 1 (dogmatic prior)
  - the posterior is always more precise than both the prior and the data
- Marginal likelihood: integrating β out gives $\hat{\beta} \sim N(m_0, v_0 + s^2)$
  - the prior variance and the sampling variance add

Prerequisites for Today (3/22): a conjugate update and empirical Bayes



- Left: prior $N(0.5, 0.3^2)$ and data $\hat{\beta} = 0.9$, $s = 0.15$: precisions $11.1 + 44.4$, prior weight 0.20, posterior mean $0.2 \cdot 0.5 + 0.8 \cdot 0.9 = 0.82$
- Right: prior $\beta \sim N(0, \tau^2)$; $\ln p(\hat{\beta} \mid \tau^2)$ is maximal at $\hat{\tau}^2 = \max(0, \hat{\beta}^2 - s^2)$

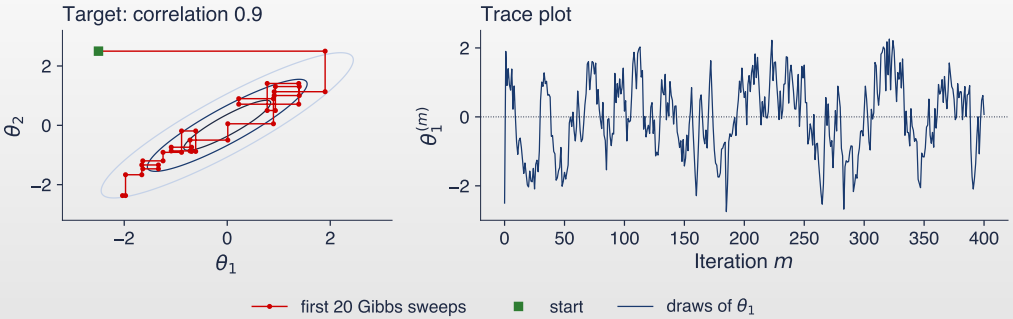
Prerequisites for Today (4/22): Gibbs sampling

- **Gibbs** Geman and Geman (1984): draw each block from its full conditional, i.e. its distribution given the other blocks and the data
  - sweep m: $\theta_1^{(m)} \sim p(\theta_1 \mid \theta_2^{(m-1)}, y)$, then $\theta_2^{(m)} \sim p(\theta_2 \mid \theta_1^{(m)}, y)$
  - the draws form a Markov chain whose limiting distribution is the joint posterior
- Regression $y = X\beta + e$, $e \sim N(0, \sigma^2 I)$: two blocks
  - $\beta \mid \sigma^2, y$ is Normal; $\sigma^2 \mid \beta, y$ is inverse gamma
- **Inverse gamma** $IG(a, d)$: $\sigma^2 \sim IG(a, d)$ if $1/\sigma^2 \sim$ Gamma with shape a and rate d

$$p(\sigma^2) \propto (\sigma^2)^{-a-1} e^{-d/\sigma^2}, \qquad \mathbb{E}\sigma^2 = \frac{d}{a-1} \ (a > 1)$$

  - conjugate for a variance: data with T residuals e update $a \to a + T/2$, $d \to d + e'e/2$

Prerequisites for Today (5/22): an example of a Gibbs sampler



- Target: two standard Normal parameters with correlation 0.9; each conditional is $N(0.9\,\theta_{\mathbf{other}}, 1 - 0.9^{\mathbf{2}})$
- Left: moves parallel to the axes, small steps along the diagonal; right: the chain of $\theta_{\mathbf{1}}$ is strongly autocorrelated (lag-1 correlation 0.81)

Prerequisites for Today (6/22): MCMC diagnostics

- **Effective sample size**: the number of independent draws worth the same as M correlated draws

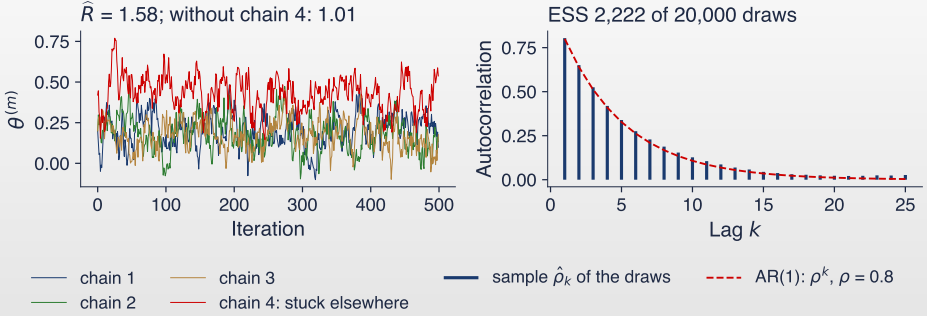
$$\text{ESS} = \frac{M}{1 + 2\sum_{k \geq 1} \rho_k}$$

  - ρ_k: autocorrelation of the draws at lag k; $1 + 2\sum_k \rho_k$: the inefficiency factor
  - AR(1) chain, $\rho_k = \rho^k$: $\text{ESS} = M(1-\rho)/(1+\rho)$
- **Potential scale reduction** Gelman and Rubin (1992): C chains of n draws each

$$\widehat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}, \qquad B = \frac{n}{C-1} \sum_{c=1}^{C} (\bar{\theta}_c - \bar{\theta})^2$$

  - W: the average within-chain variance; $\bar{\theta}_c$: the mean of chain c; $\bar{\theta}$: the overall mean
  - $\widehat{R} \geq 1$; close to 1 (below 1.01) and a large ESS are necessary, not sufficient, for convergence

Prerequisites for Today (7/22): trace plots, $\widehat{R}$ and the ESS



- Left: chain 4 explores another region: the chain means differ, B is large and $\widehat{R} = 1.58$; without it, $\widehat{R} = 1.01$
- Right: a chain with $\rho = 0.8$; 20 000 draws are worth $20\,000 \cdot 0.2/1.8 = 2\,222$ independent draws

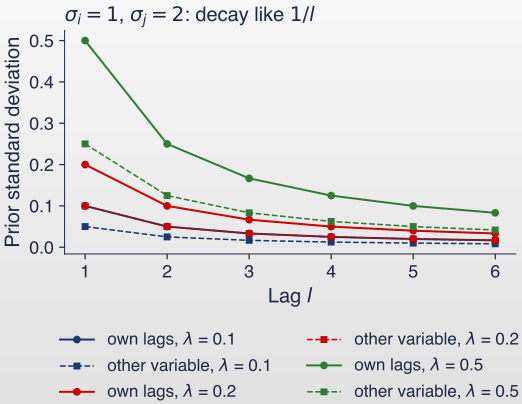
Prerequisites for Today (8/22): the VAR as a regression

- VAR(p) with n variables

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \sim N(0, \Sigma)$$

- Stacked as one multivariate regression $Y = XB + U$
  - Y ($T \times n$): the rows y_t'; X ($T \times k$): the rows $(1, y_{t-1}', \ldots, y_{t-p}')$, $k = 1 + np$
  - $B = (c, A_1, \ldots, A_p)'$ ($k \times n$); U: the errors; $(A_l)_{ij}$: the effect of variable j at lag l on variable i
- OLS: $\hat{B} = (X'X)^{-1}X'Y$, equation by equation
  - $n = 6$, $p = 12$: $k = 73$ coefficients per equation; with about 120 observations OLS fits noise
  - shrinkage towards a simple prior model trades a little bias for much less variance

Prerequisites for Today (9/22): the Minnesota prior



- Prior means Litterman (1986): each variable a random walk
  - $\mathbb{E}(A_{\mathbf{1}})_{ii} = \delta_i$ ($\delta_i = 1$, or 0 for stationary series), all other coefficients 0
- Prior standard deviations of $(A_l)_{ij}$

$$\frac{\lambda}{l} \quad (i = j), \qquad \frac{\lambda}{l}\frac{\sigma_i}{\sigma_j} \quad (i \neq j)$$

  - λ: overall tightness; small λ: strong shrinkage
  - $1/l$: distant lags are shrunk more
  - σ_i^2: residual variance of an AR for y_i; σ_i/σ_j fixes the units

Prerequisites for Today (10/22): the natural conjugate prior

- Normal–inverse Wishart Kadiyala and Karlsson (1997)

$$\mathrm{vec}B \mid \Sigma \sim N(\mathrm{vec}B_0,\ \Sigma \otimes \Omega_0), \qquad \Sigma \sim IW(\Psi, d)$$

  - $\mathrm{vec}B$: the columns of B stacked; B_0: prior mean; Ω_0 ($k \times k$): prior covariance across regressors
  - $\otimes$: Kronecker product, $(\Sigma \otimes \Omega_0)$ has blocks $\sigma_{ij}\Omega_0$; $IW(\Psi, d)$: inverse Wishart with scale Ψ and d degrees of freedom
  - the same Ω_0 in every equation: cross-variable shrinkage cannot differ from own shrinkage

- Posterior mean: a precision-weighted average of OLS and B_0

$$\bar{B} = (X'X + \Omega_0^{-1})^{-1}(X'Y + \Omega_0^{-1}B_0)$$

  - $\Omega_0 \to 0$: $\bar{B} \to B_0$ (the prior); $\Omega_0^{-1} \to 0$: $\bar{B} \to \hat{B}$ (OLS); a ridge regression towards B_0

Prerequisites for Today (11/22): dummy observations

- **Dummy observations** Bańbura, Giannone and Reichlin (2010): artificial rows (Y_d, X_d) that state the prior as if it were data
  - OLS on the stacked data equals the posterior mean

$$\bar{B} = (X_d'X_d + X'X)^{-1}(X_d'Y_d + X'Y), \quad \Omega_0 = (X_d'X_d)^{-1}, \; B_0 = \Omega_0 X_d' Y_d$$

  - one coefficient with prior $N(\delta, \lambda^2)$ and $\sigma = 1$: one row $y_d = \delta/\lambda$, $x_d = 1/\lambda$
- **Sum of coefficients** Doan, Litterman and Sims (1984): n rows $Y_d = \mathrm{diag}(\bar{y})/\mu$, $X_d = (0, 1_p' \otimes \mathrm{diag}(\bar{y})/\mu)$
  - $\bar{y}$: the means of the first p observations; 1_p: a vector of p ones; 0: the constant column
  - $\mu \to 0$: $I - \sum_l A_l \to 0$, i.e. unit roots and no cointegration (a VAR in differences)
- **Initial observation** Sims and Zha (1998): one row $(1/\phi, 1_p' \otimes \bar{y}'/\phi)$ against $\bar{y}'/\phi$
  - $\phi \to 0$: unit roots or a stationary mean equal to $\bar{y}$; allows cointegration

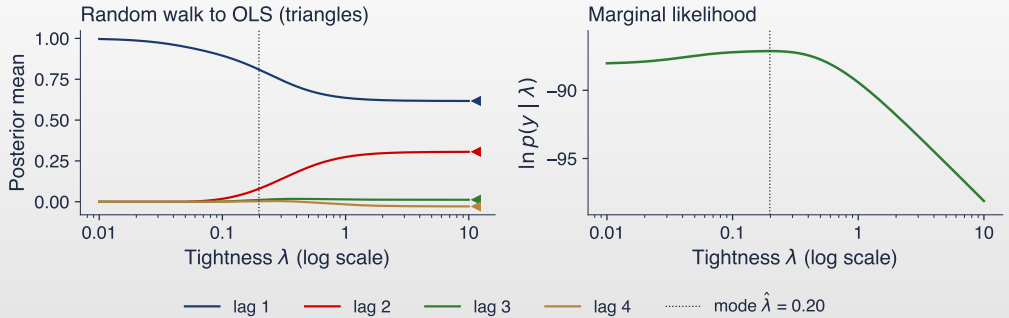
Prerequisites for Today (12/22): hyperparameters by the marginal likelihood

- GLP Giannone, Lenza and Primiceri (2015): treat (λ, μ, ϕ) as parameters with their own prior (a hyperprior)

$$p(\lambda, \mu, \phi \mid Y) \propto p(Y \mid \lambda, \mu, \phi)\, p(\lambda)\, p(\mu)\, p(\phi)$$

  - $p(Y \mid \lambda, \mu, \phi)$: the marginal likelihood of the BVAR, in closed form for the natural conjugate prior
  - $p(\cdot)$: Gamma hyperpriors, with mode 0.2 for λ and 1 for μ and ϕ
  - the posterior mode of (λ, μ, ϕ) is then plugged into the BVAR
- Reading the mode
  - a small $\hat{\lambda}$: the data prefer strong shrinkage; it decreases as the number of variables grows Bańbura, Giannone and Reichlin (2010)
  - small $\hat{\mu}$, $\hat{\phi}$: the data favour persistence and near unit roots

Prerequisites for Today (13/22): shrinkage and the marginal likelihood



- Simulated AR(4) with $T = 60$, random-walk prior (lag 1 at 1, other lags at 0), standard deviation λ/l
- Left: as λ grows the posterior means move from the prior to OLS; right: $\ln p(y \mid \lambda)$ peaks at $\hat{\lambda} = 0.20$

Prerequisites for Today (14/22): forecast evaluation by RMSE

- Recursive forecasts: re-estimate at each origin τ with data up to τ, forecast h steps ahead
  - $e_{\tau+h} = y_{\tau+h} - \hat{y}_{\tau+h|\tau}$: the forecast error; N_o: the number of origins
- Root mean squared error, in the units of y

$$\mathrm{RMSE}_h = \sqrt{\frac{1}{N_o} \sum_{\tau} e_{\tau+h}^2}$$

- Benchmark: random walk with drift, $\hat{y}_{\tau+h|\tau} = y_\tau + h\hat{g}$
  - $\hat{g}$: the mean change up to τ; a model is useful only if its RMSE is below that of the benchmark
  - with few origins the RMSE differences are noisy (a test, Diebold–Mariano, is in Chapter 1)

Prerequisites for Today (15/22): factor models and principal components

- Approximate factor model: each series = common part + idiosyncratic part

$$x_{it} = \lambda_i' F_t + e_{it}, \qquad X = F\Lambda' + E$$

  - x_{it}: series i at time t, standardised (N series, T periods); F_t: the r factors; λ_i: the loadings of series i
  - e_{it}: the idiosyncratic error, weakly correlated across i
- Principal components Stock and Watson (2002a)

$$\hat{F} = \sqrt{T} \times (\text{first } r \text{ eigenvectors of } XX'), \qquad \hat{\Lambda} = X'\hat{F}/T$$

  - X ($T \times N$): the panel; $\hat{F}'\hat{F}/T = I_r$: the normalisation
  - share of variance of component j: $\mu_j / \sum_i \mu_i$, μ_j the j-th eigenvalue of the correlation matrix
  - only the factor space is identified: $F\Lambda' = (FH)(\Lambda H^{-1\prime})'$ for any invertible H
- Pervasive factors: the first r eigenvalues grow like N, the others stay bounded

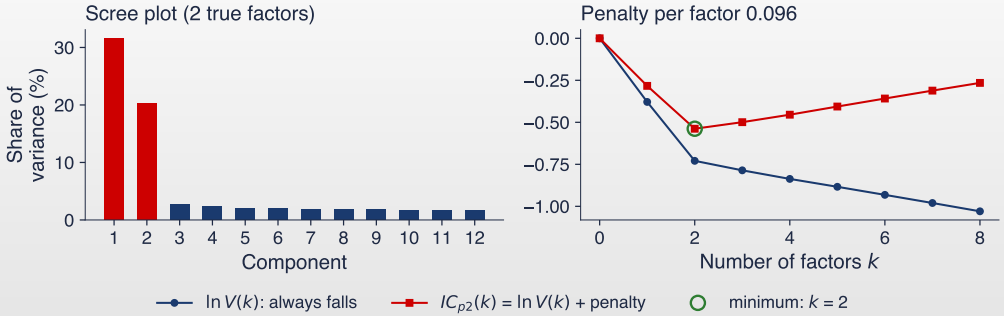
Prerequisites for Today (16/22): the number of factors

- Bai–Ng criteria Bai and Ng (2002): fit plus a penalty per factor

$$IC_{p2}(k) = \ln V(k) + k\,\frac{N+T}{NT}\ln\min(N,T), \qquad PC_{p2}(k) = V(k) + k\,\hat{\sigma}^2\,\frac{N+T}{NT}\ln\min(N,T)$$

- Notation
  - $V(k) = (NT)^{-1}\sum_{i,t}(x_{it} - \hat{\lambda}_i'\hat{F}_t)^2$: the residual mean square after k components
  - $\hat{\sigma}^2 = V(k_{\max})$: a scale for the PC penalty
- Choose $\hat{k} = \arg\min_{k \le k_{\max}}$ of the criterion
  - $V(k)$ alone never increases with k: it would always choose $k_{\max}$
  - the penalty goes to 0 but more slowly than the estimation noise of an extra factor

Prerequisites for Today (17/22): a scree plot and IC_{p2}



- Simulated panel, $N = 60$, $T = 150$, two factors: the first two components explain 32% and 20%, the others about 2% each
- $\ln V(k)$ falls at every k; with the penalty 0.096 per factor, IC_{p2} is minimal at $k = 2$

Prerequisites for Today (18/22): FAVAR

- **FAVAR** Bernanke, Boivin and Eliasz (2005): a VAR in the estimated factors and the policy rate Y_t

$$\begin{pmatrix} \hat{F}_t \\ Y_t \end{pmatrix} = \sum_{l=1}^{p} \Phi_l \begin{pmatrix} \hat{F}_{t-l} \\ Y_{t-l} \end{pmatrix} + v_t, \qquad x_{it} = \lambda_i' \hat{F}_t + \gamma_i Y_t + e_{it}$$

  - Φ_l: VAR coefficients; v_t: VAR errors; K: the number of factors in $\hat{F}_t$
  - two steps: principal components of the panel, then factors purged of Y_t using the slow-moving series
- Identification: Y_t ordered last (Cholesky), so a policy shock does not move the factors on impact
  - response of any series: $\partial x_{i,t+h} / \partial \varepsilon_t^Y = \lambda_i' \Psi_h^F + \gamma_i \Psi_h^Y$
  - Ψ_h^F, Ψ_h^Y: the responses of $\hat{F}$ and Y after h periods; price puzzle: prices rising after a tightening

Prerequisites for Today (19/22): mixed frequencies

- Quarterly growth from a latent monthly growth $y_t = \Delta \ln Y_t^M$ Mariano and Murasawa (2003)

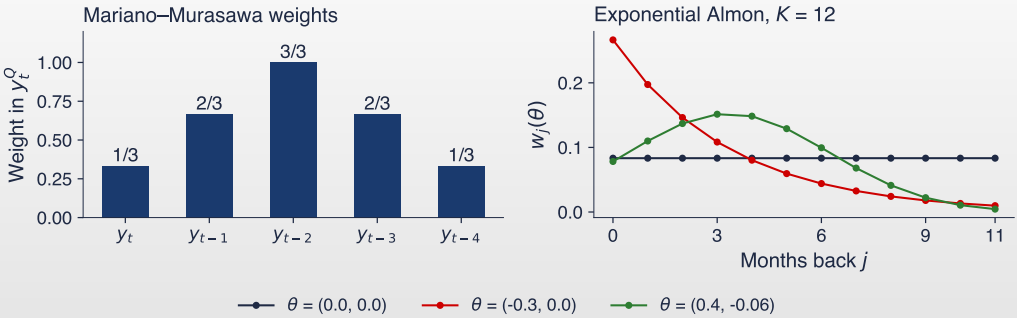
$$y_t^Q \approx \tfrac{1}{3}\left(y_t + 2y_{t-1} + 3y_{t-2} + 2y_{t-3} + y_{t-4}\right)$$

  - y_t^Q: quarter-on-quarter growth of GDP, observed in the third month of each quarter
  - it uses the log of the quarterly level $\approx$ the mean of the three monthly logs
- Bridge equation Baffigi, Golinelli and Parigi (2004): complete the quarter with AR forecasts of the monthly indicator, average to a quarter, regress
  - $y_\tau^Q = a + b\,\bar{x}_\tau + \varepsilon_\tau$, $\bar{x}_\tau$: the quarterly average of the (completed) indicator
- **MIDAS** Ghysels, Santa-Clara and Valkanov (2006): monthly lags with parsimonious weights

$$y_\tau = \beta_0 + \beta_1 \sum_{j=0}^{K-1} w_j(\theta)\, x_{m_\tau - j} + \varepsilon_\tau, \qquad w_j(\theta) = \frac{e^{\theta_1 j + \theta_2 j^2}}{\sum_{i=0}^{K-1} e^{\theta_1 i + \theta_2 i^2}}$$

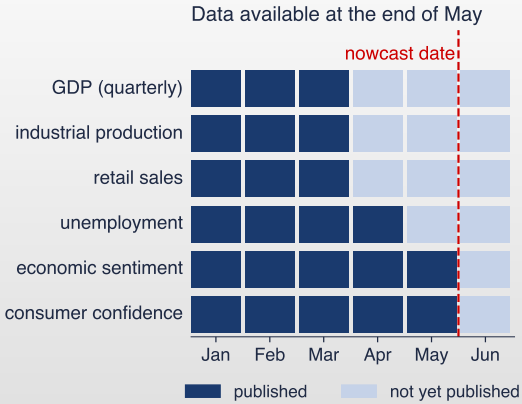
  - τ: the quarter; m_τ: its latest available month ($j = 0$); K: the number of monthly values; w_j sum to 1
  - one regression per information set (position in the quarter)

Prerequisites for Today (20/22): two weighting schemes



- Left: the Mariano–Murasawa weights $(1, 2, 3, 2, 1)/3$ sum to 3, because quarterly growth spans five months of monthly growth
- Right: $\theta = (0, 0)$ gives equal weights, $\theta_1 < 0$ declining weights, $\theta_2 < 0 < \theta_1$ a hump

Prerequisites for Today (21/22): the ragged edge and the DFM

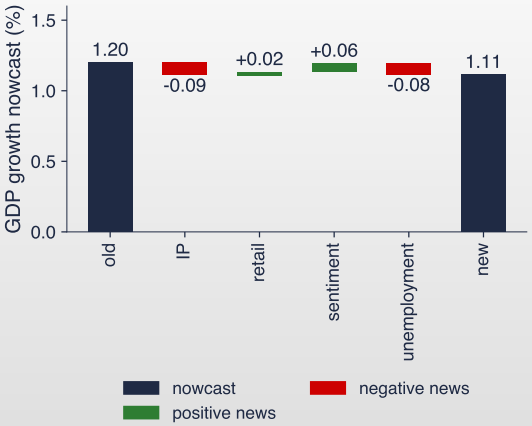


- Ragged edge: series end in different months
  - surveys: no delay; unemployment: 1 month; hard data: 2 months; GDP: 2 months after the quarter
- **DFM** (dynamic factor model) in state space

$$x_t = \Lambda f_t + e_t, \qquad f_t = \sum_{l=1}^{p} \Phi_l f_{t-l} + \eta_t$$

  - f_t: the latent factors, following a VAR
  - the Kalman filter skips missing x_{it}: no balancing of the panel
  - GDP enters through the Mariano–Murasawa weights

Prerequisites for Today (22/22): news and nowcast revisions



- News Bańbura and Modugno (2014): the unexpected part of a release
  - $I_j = x_j - \mathbb{E}[x_j \mid \Omega_{old}]$; Ω_{old}: information before the release
- Revision = weighted sum of the news

$$\mathbb{E}[y \mid \Omega_{new}] - \mathbb{E}[y \mid \Omega_{old}] = \sum_j w_j I_j$$

  - $w = \mathbb{E}[yI'] \, \mathbb{E}[II']^{-1}$: regression of y on the news
  - an expected release ($I_j = 0$) changes nothing
- Chart (illustrative): IP below expectations (-0.09) and unemployment above (-0.08) outweigh sentiment ($+0.06$): $1.20 \to 1.11$

A1: conjugate updating and empirical Bayes [Solved]

Task

- AR(1) without constant, $\sigma^2 = 1$ known, $\sum x_t^2 = 50$, $\sum x_t y_t = 40$; prior $\rho \sim N(1, 0.2^2)$.
- 1. Compute $\hat{\rho}$ and its standard error s.
- 2. Compute the posterior mean, standard deviation and the weight of the prior.
- 3. With the prior $\rho \sim N(0, \tau^2)$, find the τ^2 that maximises the marginal likelihood of $\hat{\rho}$.
- Report: two numbers, three numbers, one number.

Solution

- 1. $\hat{\rho} = 40/50 = 0.800$, $s = 1/\sqrt{50} = 0.141$
- 2. Precisions $25 + 50 = 75$: $m_1 = (25 \cdot 1 + 50 \cdot 0.8)/75 = 0.8667$, sd $= 1/\sqrt{75} = 0.1155$, prior weight $25/75 = 0.333$
- 3. $\hat{\rho} \sim N(0, \tau^2 + s^2)$; maximising in τ^2: $\tau^2 + s^2 = \hat{\rho}^2$, so $\hat{\tau}^2 = \max(0, \hat{\rho}^2 - s^2) = 0.64 - 0.02 = 0.620$
- Interpretation: the data choose how much to shrink; this is the one-parameter version of GLP

Back Exercise list

A2: Minnesota prior variances and a ridge posterior [Proposed]

Solution discussed in class

Task

- Bivariate VAR(2), equation 1; $\lambda = 0.2$, $\sigma_{\mathbf{1}} = 1$, $\sigma_{\mathbf{2}} = 2$, $\delta_{\mathbf{1}} = 1$. Model: A1.
- 1. Give the prior standard deviations of the own and cross coefficients at lags 1 and 2.
- 2. Keep only $y_{\mathbf{1},t-\mathbf{1}}$ and $y_{\mathbf{2},t-\mathbf{1}}$: $X'X = \begin{pmatrix} 100 & 20 \\ 20 & 400 \end{pmatrix}$, $X'y = (85, 30)'$, $\sigma^{\mathbf{2}} = 1$; compute the OLS and the posterior means.
- 3. Say what happens as $\lambda \to 0$ and as $\lambda \to \infty$.
- Report: four numbers, two pairs of numbers, two sentences.

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A3: Gibbs conditionals and MCMC diagnostics [Solved]

Task

- $y = X\beta + e$, $e \sim N(0, \sigma^2 I)$, priors $\beta \sim N(b_0, V_0)$, $\sigma^2 \sim IG(a_0, d_0)$, independent.
- 1. Derive $\beta \mid \sigma^2, y$ and $\sigma^2 \mid \beta, y$.
- 2. A chain of $M = 10\,000$ draws has AR(1) autocorrelation $\rho = 0.9$: compute the ESS.
- 3. Four chains of $n = 1\,000$ draws have means $(0.10, 0.12, 0.35, 0.11)$ and within variance $W = 0.01$: compute $\widehat{R}$.
- Report: two densities, one number, one number and a verdict.

Solution

- 1. $\beta \mid \cdot \sim N(\bar{b}, \bar{V})$, $\bar{V} = (V_0^{-1} + X'X/\sigma^2)^{-1}$, $\bar{b} = \bar{V}(V_0^{-1} b_0 + X'y/\sigma^2)$; $\sigma^2 \mid \cdot \sim IG(a_0 + T/2, d_0 + e'e/2)$
- 2. $\sum_k \rho^k = \rho/(1-\rho)$: $\text{ESS} = M(1-\rho)/(1+\rho) = 526.3$ (one effective draw every 19)
- 3. $B = n\,\mathrm{var}(\bar{\theta}_c) = 14.467$, $\hat{V} = 0.999W + B/n = 0.0245$, $\widehat{R} = 1.56$: not converged; chain 3 is stuck elsewhere (without it, $\widehat{R} = 1.004$)

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A4: dummy observations [Proposed]

Solution discussed in class

Task

- The AR(1) of A1; Minnesota dummy $y_d = \sigma\delta/\lambda$, $x_d = \sigma/\lambda$ with $\sigma = 1$, $\delta = 1$, $\lambda = 0.2$. Model: A1, A3.
- 1. Show that OLS on the augmented data equals $(X'X + X_d'X_d)^{-\mathbf{1}}(X'y + X_d'y_d)$.
- 2. Compute it and compare with A1.
- 3. Add a sum-of-coefficients dummy $y = x = \bar{y}/\mu$ with $\bar{y} = 5$, $\mu = 1$ and recompute; say where it pushes ρ.
- Report: one derivation, one number, one number and one sentence.

A5: principal components of an equicorrelated panel [Solved]

Task

- $N = 3$ standardised series with all pairwise correlations $\rho = 0.6$.
- 1. Find the eigenvalues and eigenvectors of the correlation matrix R.
- 2. Compute the share of variance of the first component and its loadings.
- 3. Give the share for $N = 100$ and its limit as $N \to \infty$.
- Report: three eigenvalues, one share, one share and a limit.

Solution

- 1. $R = (1 - \rho)I + \rho 11'$: $\lambda_{\mathbf{1}} = 1 + 2\rho = 2.2$ with $1/\sqrt{3}$; $\lambda_{\mathbf{2,3}} = 1 - \rho = 0.4$
- 2. Share $2.2/3 = 73.3\%$; loadings $1/\sqrt{3} = 0.577$ each: the factor is the cross-sectional average
- 3. $(1 + 99\rho)/100 = 60.4\%$, limit ρ: the first eigenvalue grows like N, the others stay bounded (pervasiveness)

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A6: the number of factors by Bai and Ng [Proposed]

Solution discussed in class

Task

- $N = 100$, $T = 200$; $V(k)$ for $k = 0, \ldots, 5$: 1.000, 0.620, 0.480, 0.430, 0.415, 0.405. Model: A5.
- 1. Compute the IC_{p2} penalty per factor.
- 2. Compute $IC_{p2}(k)$ for each k and choose k.
- 3. Explain why $V(k)$ alone always picks $k = 5$, and why the chosen factors are defined only up to a rotation.
- Report: one number, a table of six values and k, two sentences.

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A7: the Mariano–Murasawa weights [Solved]

Task

- Quarterly GDP is the 3-month average of a latent monthly level Y_t^M; $y_t = \Delta \ln Y_t^M$.
- 1. Approximate $\ln Y_t^Q$ by the average of the three monthly logs.
- 2. Derive $y_t^Q = \ln Y_t^Q - \ln Y_{t-\mathbf{3}}^Q$ as a combination of $y_t, \ldots, y_{t-\mathbf{4}}$.
- 3. With $(y_t, \ldots, y_{t-\mathbf{4}}) = (0.4, 0.3, -0.1, 0.5, 0.2)$, compute y_t^Q.
- Report: one formula, five weights, one number.

Solution

- 1. Geometric mean instead of the arithmetic mean: $\ln Y_t^Q \approx \frac{\mathbf{1}}{\mathbf{3}} \sum_{j=\mathbf{0}}^{\mathbf{2}} \ln Y_{t-j}^M$
- 2. Write each log difference over three months as a sum of monthly growth rates: weights $(1, 2, 3, 2, 1)/3$, summing to 3
- 3. $(0.4 + 0.6 - 0.3 + 1.0 + 0.2)/3 = 0.633$

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A8: news and MIDAS weights [Proposed]

Solution discussed in class

Task

- Standardised y (GDP growth) and survey release x, $\mathrm{Cov}(y, x) = 0.6$, $\mathrm{Var}x = 1$; $\mathbb{E}[x \mid \Omega_{old}] = 0.2$, released $x = -0.8$. Model: A7.
- 1. Compute the news and the revision of the nowcast.
- 2. Two releases with $\mathbb{E}[II'] = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, $\mathbb{E}[yI] = (0.6, 0.3)'$: compute the weights and say why the second is zero.
- 3. Exponential Almon with $\theta = (0.5, -0.2)$, $K = 4$: compute the weights.
- Report: two numbers, two weights and a sentence, four weights.

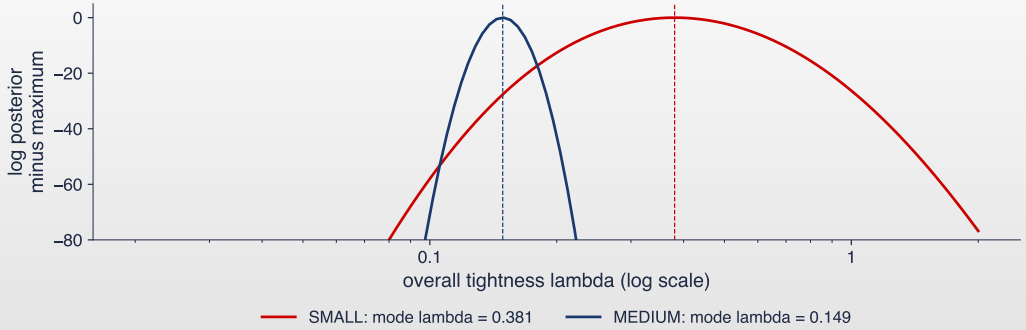
Back Exercise list

B1: hyperparameters by marginal likelihood [Solved]

- **Question**: how tight should the Minnesota prior be for the US SMALL and MEDIUM systems?
- **Inputs**: SMALL (private employment, CPI, funds rate) and MEDIUM (20 series of BGR), 100 $\times$ log levels or levels, $p = 13$, 1960:1–2019:12
- **Tasks**
  1. Compute the posterior mode of (λ, μ, ϕ) with the GLP Gamma hyperpriors for each system.
  2. Plot the log posterior as a function of λ and report its loss at $\lambda = 0.2$ and $\lambda = 1$.
  3. Compute the sum of the own-lag coefficients of CPI in the posterior mean (and by OLS for SMALL).
  4. Interpretation: what do the modes say about the size of the system and about unit roots?
- **Report**: six hyperparameters, four losses, three sums, one sentence
- Notebook: section B1

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B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

System	λ	μ	ϕ	loss at 0.2 / 1	CPI own sum
SMALL ($k = 40$)	0.381	0.199	0.440	-12.6 / -26.3	0.9999 (OLS 0.9980)
MEDIUM ($k = 261$)	0.149	0.134	0.445	-42.6 / -2173.5	1.0000

- Interpretation: the larger system wants a tighter prior
  - λ 0.381 against 0.149; the BGR fit rule gives 0.158
  - small μ and ϕ: the data favour persistence and near unit roots
  - the CPI own-lag sum is almost exactly 1

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B2: a Bayesian VAR for Romania [Proposed]

- **Question**: does a GLP-type BVAR forecast Romanian inflation and industrial production better than a random walk and an OLS VAR?
- **Inputs**: monthly 2005:8–2026:7 ($T = 252$): 100 log IP, 100 log HICP, ROBOR 3M, 100 log EUR/RON (BNR), unemployment, ESI; $p = 12$; model: B1
- **Tasks**
  1. Set $\delta_i = 1$ for all variables except the ESI and compute the GLP mode on the full sample.
  2. Forecast recursively from origins 2015:1–2025:7 the 1- and 12-month changes of log HICP and log IP with RW with drift, OLS VAR(12) and the BVAR (hyperparameters re-optimised each January).
  3. Report the RMSE table and the range of the re-optimised λ.
  4. Interpretation: what does the OLS VAR teach about $k = 73$ regressors and 120 observations?
- **Report**: one set of hyperparameters, a table of 12 RMSEs, one interval, one sentence
- Notebook: section B2

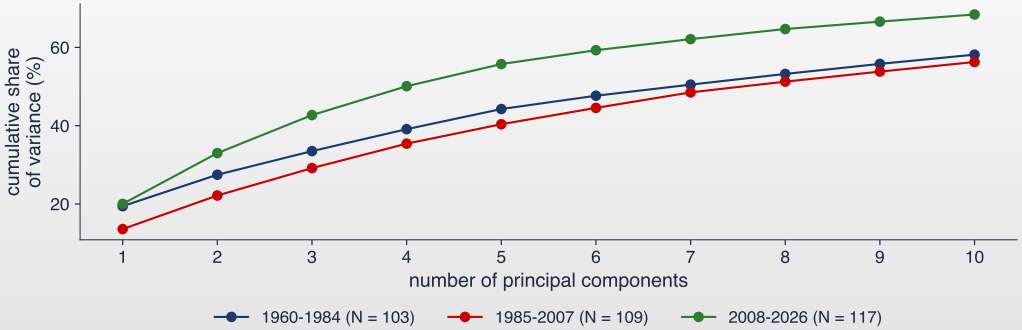
Back Exercise list

B3: factors of the FRED-MD panel on three samples [Solved]

- **Question**: is the factor structure of the US economy stable over time?
- **Inputs**: FRED-MD-format panel, transformed and cleaned of outliers; samples 1960–1984, 1985–2007, 2008–2026 (series with at least 80% of observations)
- **Tasks**
  1. Estimate eight factors by EM principal components on each sample.
  2. Report the Bai–Ng choices (IC_{p2}, PC_{p2}), the share of the first component and of the first three.
  3. Report the absolute correlation of the first factor with IP growth, CPI inflation (second difference) and the change of the funds rate.
  4. Interpretation: what changed across the three periods?
- **Report**: a table of three rows and seven columns, one sentence
- Notebook: section B3

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B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

Sample	N	IC_{p2}	PC_{p2}	1st, %	3, %	IP	CPI	FFR
1960–1984	103	6	10	19.4	33.5	0.85	0.10	0.40
1985–2007	109	7	10	13.6	29.2	0.77	0.01	0.54
2008–2026	117	8	10	20.0	42.7	0.80	0.29	0.42

- Interpretation: the first factor is real activity in all periods
  - it explained least in the Great Moderation and most after 2008 (financial crisis, 2020)
  - the criteria choose six to eight factors, more in the recent sample

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B4: FAVAR and the number of factors [Proposed]

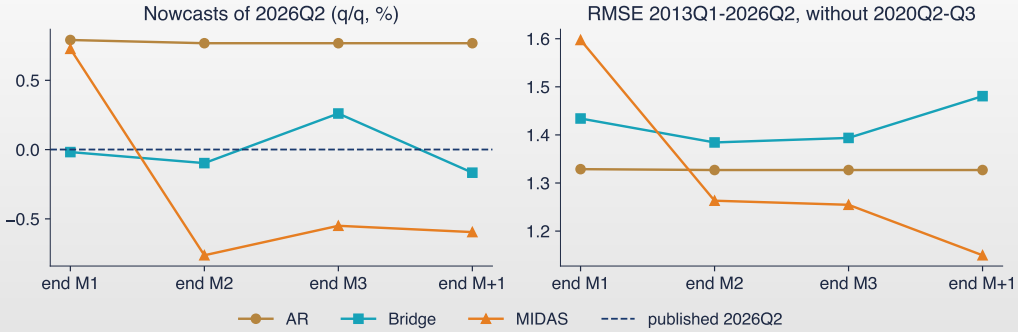
- **Question**: do the FAVAR responses to a monetary policy shock depend on the number of factors K?
- **Inputs**: balanced FRED-MD-format panel 1960:1–2001:8, two-step FAVAR of Bernanke, Boivin and Eliasz (2005), VAR(13), funds rate last, 25 bp shock; model: B3
- **Tasks**
  1. Estimate the FAVAR with $K = 1$, 3 and 5.
  2. Report the IP trough and its month, the largest CPI response within 24 months, the CPI response after 48 months and the unemployment peak.
  3. Interpretation: which K would you report, and how would you defend it before seeing the responses?
- **Report**: a table of three rows and five columns, two sentences
- Notebook: section B4

B5: nowcasting Romanian GDP in 2026Q2 [Solved]

- **Question**: how did the AR, bridge and MIDAS nowcasts of 2026Q2 evolve as the data arrived, and how accurate are they in general?
- **Inputs**: monthly panel (Eurostat) with stylised release lags (surveys 0, unemployment 1, hard data 2 months; GDP 2 months after the quarter); published growth 0.00%
- **Tasks**
  1. Build the information sets at the ends of April, May, June and July 2026.
  2. Compute the AR(1), bridge (IP, retail, ESI) and MIDAS (IP, ESI, $K = 6$) nowcasts at each date.
  3. Compute the RMSE of the same models for 2013Q1–2026Q2 without 2020Q2–Q3.
  4. Interpretation: is one quarter enough to rank models?
- **Report**: a table of twelve nowcasts, a table of twelve RMSEs, one sentence
- Notebook: section B5

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B5: solution (1/2) [Solved]



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B5: solution (2/2) [Solved]

End of	AR	Bridge	MIDAS	RMSE AR	RMSE bridge	RMSE MIDAS
April	0.79	−0.02	0.73	1.33	1.43	1.60
May	0.77	−0.10	−0.76	1.33	1.38	1.26
June	0.77	0.26	−0.55	1.33	1.39	1.25
July	0.77	−0.17	−0.60	1.33	1.48	1.15

- Interpretation: one quarter says almost nothing about a ranking
  - 2026Q2: the bridge nowcast was closest to the published 0.00%, MIDAS overshot downwards
  - over 2013–2026, MIDAS is best from month 2 on and the bridge worst

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B6: the 2026Q3 nowcast and its news [Proposed]

- **Question**: how did the September 2026 releases change the dynamic-factor nowcast of 2026Q3?
- **Inputs**: the panel of B5; two-factor DFM (two-step, Mariano–Murasawa link); information sets at the ends of August and September 2026; model: B5
- **Tasks**
  1. Estimate the DFM on the August information set and compute the nowcast of 2026Q3.
  2. Keeping the parameters fixed, compute the September nowcast and the news decomposition by series.
  3. Check that the impacts add up to the revision; compare with the EM estimate of statsmodels.
  4. Interpretation: which releases moved the nowcast, hard or soft data?
- **Report**: two nowcasts, a table of impacts, one check, two EM numbers, one sentence
- Notebook: section B6

C1: do surveys still help after 2020? [Proposed]

- **Question**: do the six Romanian confidence balances improve the end-of-quarter DFM nowcast beyond the hard data, before and after 2020?
- **Inputs**: the panel of B5; DFM with all series (two factors) and with hard data only (one factor); quarters 2013–2019 and 2021–2026; model: B5
- **Tasks**
  1. Write a five-line pre-registration: target, models, information set, sample split, test.
  2. Compute the RMSE of both DFMs and of the AR at the end of month 3 on each subsample.
  3. Interpretation: what can and cannot be concluded from about 25 quarters per subsample?
- **Report**: a five-line plan, a table of six RMSEs, a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise Bayesian VARs, factor models and nowcasting. The answer:
- `(a) With the Minnesota prior, lambda = 0 gives the OLS estimates.`
- `(b) The natural conjugate prior allows a cross-variable shrinkage theta below 1.`
- `(c) In large BVARs the optimal lambda increases with the number of variables.`
- `(d) The sum-of-coefficients prior pushes the VAR towards a model in first differences.`
- `(e) Bai-Ng IC_p2 picks the number of factors that maximises the explained variance.`
- `(f) The news of a release is its value; its impact is the value times the loading.`
- `(g) A Gibbs sampler with R-hat = 1.56 has converged.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- A prior is information: precisions add, and the marginal likelihood tells how much to trust it
- Dummy observations turn a Bayesian VAR into OLS on augmented data
- Principal components and Bai–Ng summarise a large panel; the number of factors can change conclusions
- Nowcasts improve as data arrive; revisions decompose into news
- An AI answer is a draft: check limits, counts, signs of diagnostics and definitions

After the seminar

- Lecture 5 develops each topic of today: Minnesota and NIW priors, GLP, large BVARs, factors, FAVAR, DFM, nowcasting of Romanian GDP
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a real-time nowcasting model for Romania
- Reading: Bańbura, Giannone and Reichlin (2010); Giannone, Lenza and Primiceri (2015); Stock and Watson (2002b); Bańbura and Modugno (2014)
- **The seminar is for practice and is not graded**

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